

EDUCATION			
YEAR	DEGREE / EXAM	INSTITUTE	CGPA / MARKS
2025-27	M.Tech.	Indian Institute of Technology, Tirupati	8.83
2017-20	M.Sc.	Indian Institute of Technology, KANPUR	6.2
2014-17	B.Sc.	University of Delhi, New Delhi	76.69%

PROJECTS	
Geo-Spatial Semantic Segmentation GITHUB June-2026 Tools & Technologies: Python, PyTorch, Mask2Former, Computer Vision Defined multi-channel GeoTIFF pixel classification as a dense semantic segmentation problem and built a Mask2Former(small)-based pipeline to solve it, achieving 86.7% Mean IoU (excluding background) and 97.4% pixel accuracy across 8,973 test images. Built an end-to-end inference and georeferencing workflow, including annotation-aware visualization to make model predictions interpretable and verifiable against ground geography. Awarded Innovation Recognition, Ministry of Panchayati Raj (MoPR) Geospatial Intelligence All-India Hackathon, for the solution's applicability to real-world governance data	
Intelligent LLM Text Classification Platform GITHUB June-2026 Tools & Technologies: Python, PyTorch, GPT-2, Transformer architecture, Tiktoken, NumPy, Pandas, TensorFlow, Hugging Face Hub Modeled text generation as a next-token probability estimation problem and implemented a GPT-style transformer in PyTorch, covering tokenization through autoregressive decoding. Formulated SMS spam filtering as a binary classification problem GPT-2 (91.88% test accuracy) and benchmarked model performance against BERT (90.89% test accuracy) and RoBERTa (92.95% test accuracy) on IMDb sentiment classification.	
Enhanced YOLOv9 for Real-Time Object Detection GITHUB Apr-2026 Tools & Technologies: Python, PyTorch, YOLO-v9, Object Detection, Deep Learning Analyzed baseline detection failure modes and modeled a channel-attention mechanism (CBAM) to address them, integrating three attention modules across multi-scale feature maps of the YOLOv9 architecture. Trained and evaluated the modified architecture on PASCAL VOC (20 classes), achieving 79.2% mAP@0.5 and a 0.39% improvement in detection accuracy over the baseline. Built a diagnostic evaluation pipeline (confusion-matrix analysis, precision-recall/F1 curves) to explain and quantify the source of the improvement	
Reimagining Wikipedia in AI era GITHUB Apr-2026 Tools & Technologies: React , Express , Ollama , RAG , Vector Embeddings , Prompt Engineering , YouTube Data API Designed a retrieval-augmented generation architecture to ground free-text responses in verifiable sources, addressing unexplainable/unattributed model outputs. Modeled persona-aware prompting and retrieval logic and integrated LLMs with external APIs into a full-stack application for personalized explanations, quizzes, and summaries.	

TECHNICAL SKILLS	
Programming	C++ Python
Libraries/Frameworks	Langchain Langgraph Ollama RAG Vector Embeddings Prompt Engineering PyTorch PyTorch Geometric NumPy Scikit-learn Pandas OpenCV MongoDB
Concepts	Agentic AI RAG Diffusion model transformer ViT VLM LLM GAN

ACHIEVEMENTS	
AIR 85 in IIT JAM MATHS, 2017 Gate Score 731 , AIR 871 GATE CSE 2025 AIR 459 in IIT JAM STATS, 2017 AIR- 111 in CSIR NET(Mathematics) in June 2022 Test Awarded the KIA–IITT Scholarship for academic merit at IIT Tirupati 2026. Innovation Recognition Award, Ministry of Panchayati Raj (MoPR) Geospatial Intelligence all India AI/ML Hackathon 2026.	

CERTIFICATIONS	
Nokia Workshop 2026 on AI, 6G, and Edge Cloud Convergence, exploring next-generation AI-driven network technologies and architectures (GANs) Generative Adversarial Networks - DeepLearning.AI (Coursera) Python for Machine Learning and Data Science - Jose Portilla, Pierian training (Udemy)	